POLITECNICO
MILANO 1863

Design Document

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Students&Companies - Design Document
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Contents

1 Introduction

1.1 Purpose

This document aim to provide technical information of S&C platform. This document aims to assist the development team in implementing the desired system. The document describes the system architecture and its components in detail, including interactions between them. The Design Document (DD) defines implementation, integration, and testing strategies based on priority, effort, and stakeholder impact. In general, the main different features of this document are:

- The high-level architecture
- Main components of the system
- Interfaces provided by the components
- Design patterns adopted
- Implementation, integration, and testing

1.2 Scope

Students&Companies (S&C) is a new platform that helps students to find internships, by browsing through the platform, in which they search for an internship that a company has posted, and sending their resume and initiate their application. Companies post internships allowing students to apply for those internships. Furthermore, universities monitor the application/internship of their students, and allowing students to report a complaint regarding any stage of the procedure.

1.3 Definition, Acronyms, Abbreviations

Table 1: Abbreviations

Abbreviations	**Description**

1.4 Revision History

1.5 Reference Documents

- Specification Document: "Assignment RDD AY 2024-2025.pdf"
- Course Slides

1.6 Document Structure

Section 1 Brief description of DD and introduction of purpose and scope. It also includes definitions, acronyms, and abbreviations.

Section 2 The main part of DD is this section that contains architectural design choices, including all the components, and the interfaces used for the development of the application. Also, it contains a deployment view and a runtime view. In the end, is explained the architectural patterns chosen with the other design decisions.

Section 3 Contains how should be the user interface on mobile applications.

Section 4 Contains the traceability matrix that shows which components satisfy certain requirements.

Section 5 The implementation plan, integration plan, and testing plan show how these plans are executed.

Section 6 Shows how much time is spent by each member of the group.

Section 7 Includes the reference documents.

2 Architectural Design

2.1 Overview

The architecture of this platform is structured by 3 logic layers:

1. Presentation Layer: The presentation Layer provides means of Input, allowing the users to manipulate the system Output, allowing the system to produce the results of user manipulation SAP is having Graphical User interface (SAP GUI). The SAP GUI is installed on Individual machines which act as a presentation layer.
2. Application Layer: In this layer business logic is executed. The application layer can be installed on one machine, or it can be distributed among more than one system.
3. Database Layer: The database layer holds the data. SAP supports any relational database. SAP does not provide any database. But it supports any RDBMS. The database layer must be installed on one machine or system. Major databases which are being used in SAP implementations are Oracle and DB2.

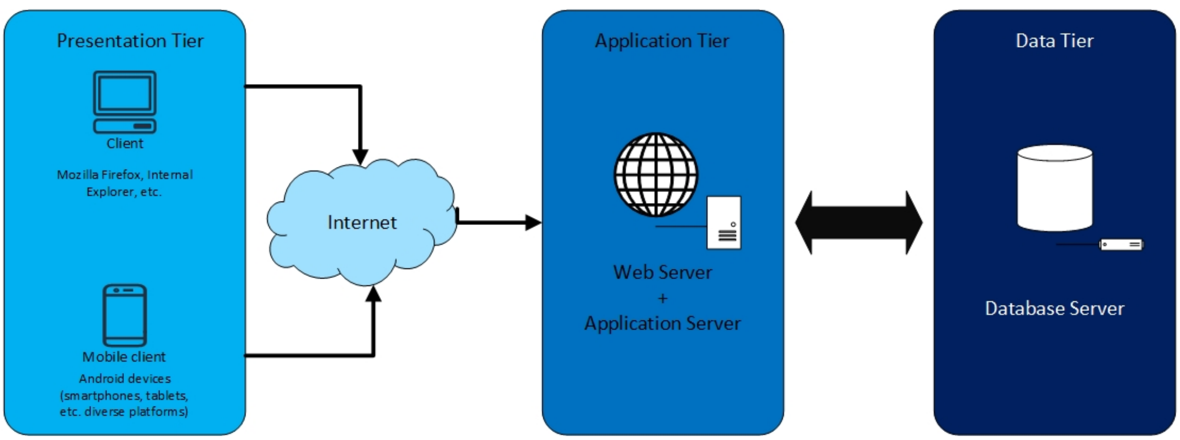

Figure 1: Three tier architecture

The following figure represents our system architecture, its high-level components, and their interaction. The client could use this by a web browser on their smartphone or computer. Then, user interactions will be processed by the application server. As well as an application server connected to the database. Finally, the application server is allowed to use APIs for Google Meets, Google Calendar.

These figures do not include all details about the architecture of the application, and in the next section, more details are represented.

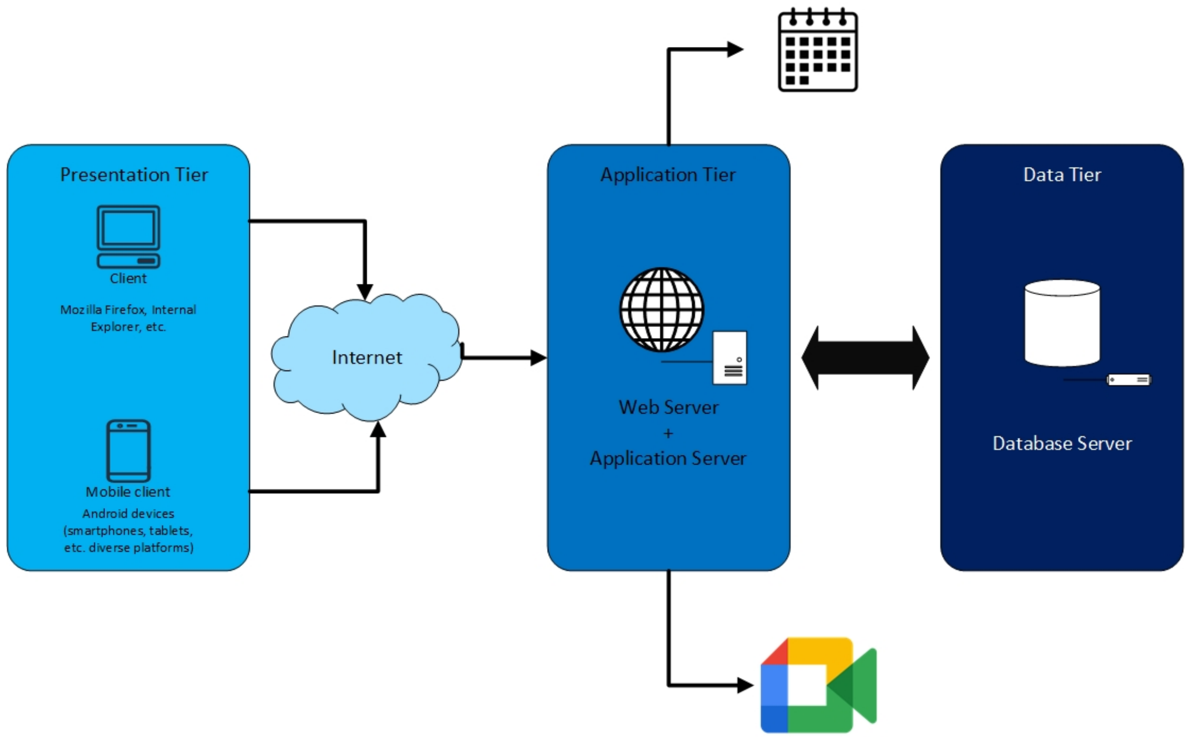

Figure 2: System Architecture

2.2 Component View

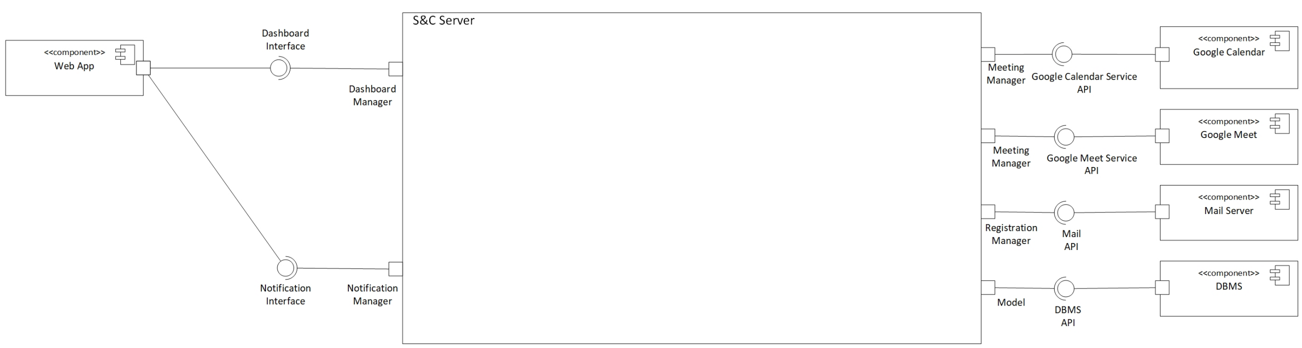

Figure 3: High Level Component Diagram

The high level component diagram above depicts S&C's external components and their communication with the server.

Web Server: The Dashboard Interface is the only way for users to communicate with the S&C Server. The S&C Server may send notifications to users via the Notification Interface, including meeting request, internship offer and etc.

DBMS: The DBMS stores all User, applications, complaints, and internships. The Model component connects to the S&C Server via the DBMS API.

Mail Server: In charge of sending emails confirming registration, the Mail Server uses the Mail API interface to connect to the S&C Server. The Registration Manager component, which manages the user registration procedure, is connected to this interface.

Google Meet: The Google Meet service is used for holding meetings in between students and company's representative for internship positions. The Google Meet service uses the Google Meet Service API interface to connect to the S&C Server. The Meeting Manager component, which manages the meetings between users, is connect to this interface.

Google Calendar: The Google Calendar service is used for keeping information about the availability regarding the meetings and internships. The Google Calendar service uses the Google Calendar API interface to connect to the S&C server. The Meeting Manager component, which manages the meetings between users, is connect to this interface.

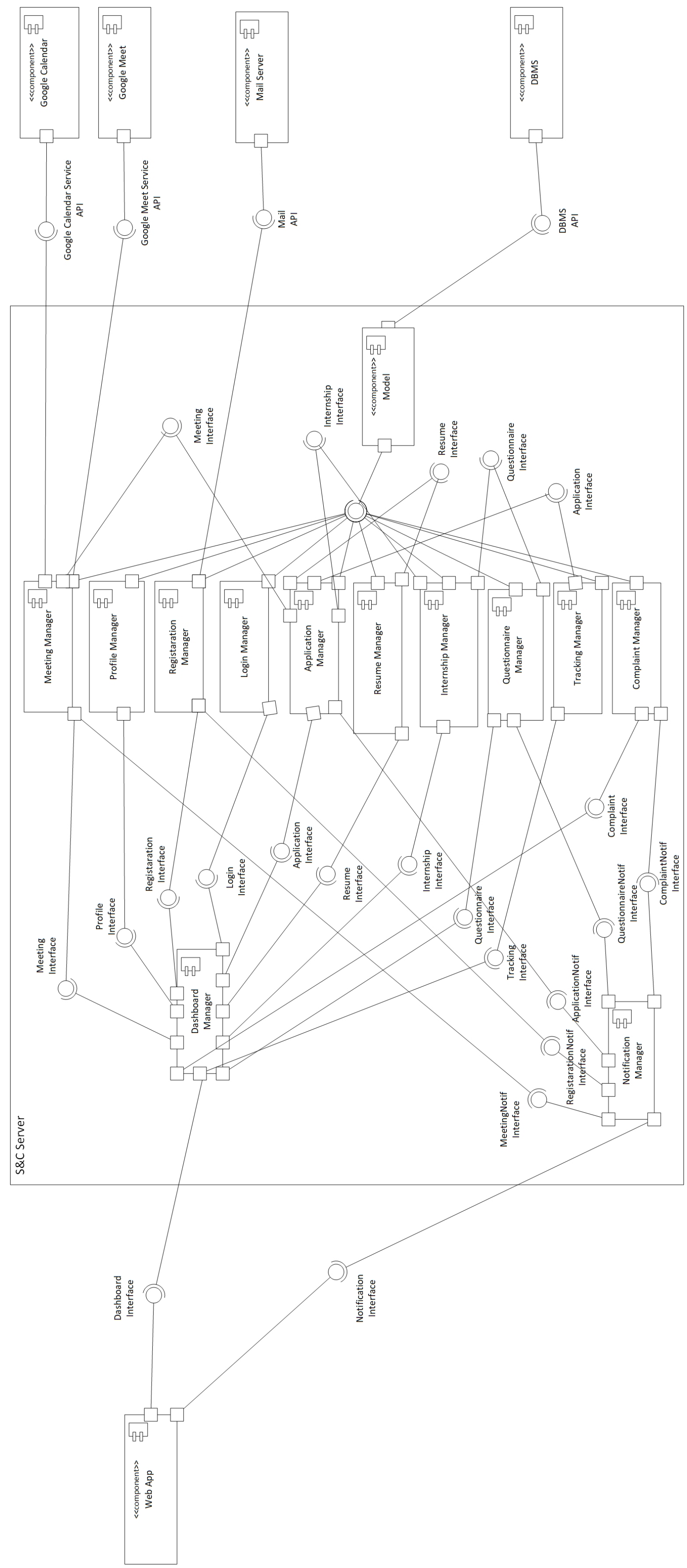

Figure 4: Low Level Component Diagram

The figure above depicts the whole architecture of the S&C website, including each component inside the S&C Server:

Dashboard Manager is a key component that facilitates communication between users and the S&C website. The Dashboard Interface allows users to interact with S&C, while the Dashboard Manager distributes requests to the necessary components. It acts as the primary center for user interactions.

The **Model** component contains an object-relation mapping (ORM) that other components use when retrieving data from the database. Thus, it is the component solely responsible for communicating with the data tier.

The **Registration Manager** is the component that manages new user registration. To create an account on the S&C system, a new user contacts with the Dashboard Manager, who then transmits the request to the Registration Manager via the registration interface. The Registration Manager sends a confirmation email to the new user and adds their information to the database using the Mail API and Model Interface. This component allows registered companies to manage internship and allows the registered students to look for internship and imitate the recruitment process.

Login Manager A component that manages the login procedure for registered users. When a user attempts to log in, the Dashboard Manager sends a request to the Login Manager via the Login Interface. The Login Manager uses the Model interface to obtain User data from the DBMS.

Notification Manager is component that handles each notification that has to be sent to the user, for example, when a new meeting is requested for an student, when user registers, any update regarding the application process for an internship, when a new questionnaire created by the a company use it send notification to all student applicants that applied for that specific internship and alerting about the new assignment, and finally we a new complaint has been submitted by an student regarding the application or internship process, a notification is sent to the university.

Application Manager handles all the users action regarding the an on-going application for an internship. It communicates with the Dashboard Manager through the Application Interface, with the Notification Manager through the ApplicationNotif Interface when a new Application is created or closed and with the Model component through the Model Interface to add or retrieve data from the DBMS.

The **Profile Manager** is responsible for managing user-specific data, including the storage, retrieval, and updating of profiles for students, companies, and universities. It provides the ability for each user type to access and modify their respective data. For example, students can update personal details or their academic credentials, while companies can update organization information or manage recruiter profiles. This component interacts with the Profile Interface to gather user inputs and provide real-time updates. It also serves as a central source of user data for other components, such as the Internship Manager, which links student profiles to internship applications, or the Login Manager, which validates user credentials.

The **Internship Manager** handles the creation, management, and tracking of internships. The main function of this component is to allow companies to post internships and manager their internship details, such as requirements, application deadlines, and enabling students to search, apply for, and track their application. This component communicates with Internship Interface, in which companies input internship details and students view available opportunities. The Internship Manager also communicates with other components such Application Manager to manage application, and Notification Manager to notify students about application updates and companies about the students actions on applications.

The **Resume Manager** is responsible for creation, storage, and management of student resumes, acting as a centralized system for handling resume related tasks. An student can upload an already built resume, or use a functionalities in the system to generate resumes directly vie Resume Interface. This component ensures that resumes are stored securely and can be retrieved for various purposes, such as applying for internships and sharing with hiring managers. It provides interface for Application Manager, which pulls the resume during internship application process.

The **Questionnaire Manager** is responsible for creation, assignment, and evaluation of the questionnaires within the system, which functions for both companies and students. Companies can use this component to design questionnaire with goal of assessing applicant's skills, knowledge, or personality as part of application procedure. Through the Questionnaire Interface, companies can create custom questionnaires and students can access and complete them. It also provides interface with internship manager, linking questionnaire with a specific internship, and the Notification Manager, which notifies users about assignment of questionnaire or submission of a questionnaire.

The **Tracking Manager** is responsible for monitoring and managing the status and progress of internship applications and related actives for students. Its main, function is to provide real-time updates on application statuses, such as submission, review, shortlisting, or rejection, ensuring transparency for all parties involved, particularly university users. Through the Tracking Interface, universities can view the progress of their student. The Tracking Manager interacts with Application Manager to retrieve application data and updates.

The **Complaint Manager** is responsible for submission, reception of complaints within the system, and enables effective communication and problem reporting among students and universities. Through the Complaint Interface, users can submit complaint regarding any issues during the application and internship procedure. Once submitted, the Complaint Manager categorize and records complaint and ensures that they are directed to appropriate university. It interacts with Notification Manager to keep users informed about the status of their complaint.

2.3 Deployment View

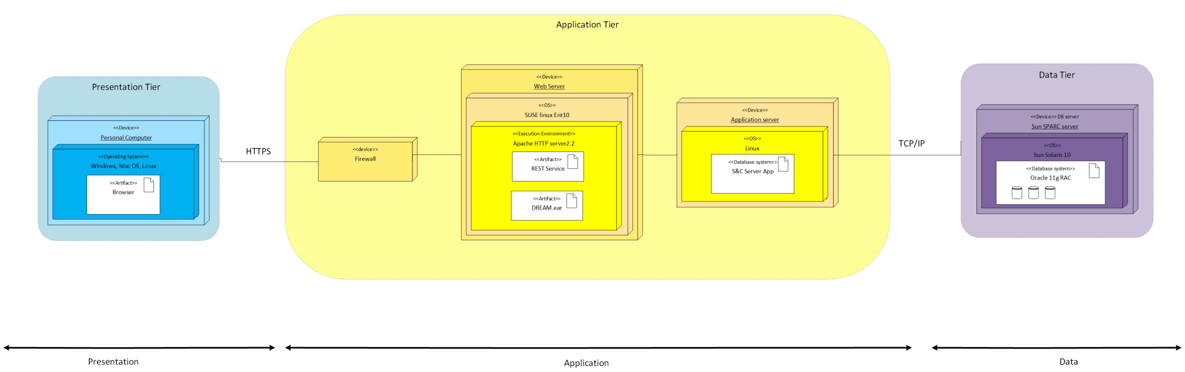

Figure 5: Deployment View Diagram

Personal Computer: Users can use their preferred web browser to access the system on any kind of personal computer. The Web Server and the browser will exchange information. Additionally, users can utilize any device that has a web browser, including tablets, smartphones, and more.

Web Server: All users that visit the system via a web browser can access the Application Server's service with the Web Server. Specifically, to manage high user traffic, the Web Server performs load balancing on client requests sent to the different Application Servers without executing any business logic. Additionally, it gives the client's browsers the HTML, JSON, Javascript, and CSS files so that the sites can be rendered.

Firewall: A firewall is used for providing safe access to the server and database, as well as, preventing external attacks.

Application Server: The system's business logic is housed in the application server. Additionally, it uses the Web Server-managed HTTPS protocol to connect with the client. The Dashboard Manager is responsible for directing the different requests that originate from the Web Server to the appropriate module. Additionally, it uses the model gateway to interface with the Database Server. To manage high user traffic, this node is duplicated.

Database Management System (DBMS): For performing the job of DBMS Oracle Real Application Cluster (RAC) has been selected. A cluster comprises multiple interconnected computers or servers that appear as if they are one server to the end-users and applications. Oracle RAC enables you to cluster Oracle databases and it is a configuration of Relation DBMS (RDBMS). Some nodes contain the same instance of the database and the instances are connected to each other and share the cache in the Global Cache (GC). Then, there is a pool of replicated databases that contain data files. Datafiles are common, shared and they are accessed from all instances in a parallel and synchrony way.

2.4 Runtime View

2.4.1 Registering as Student

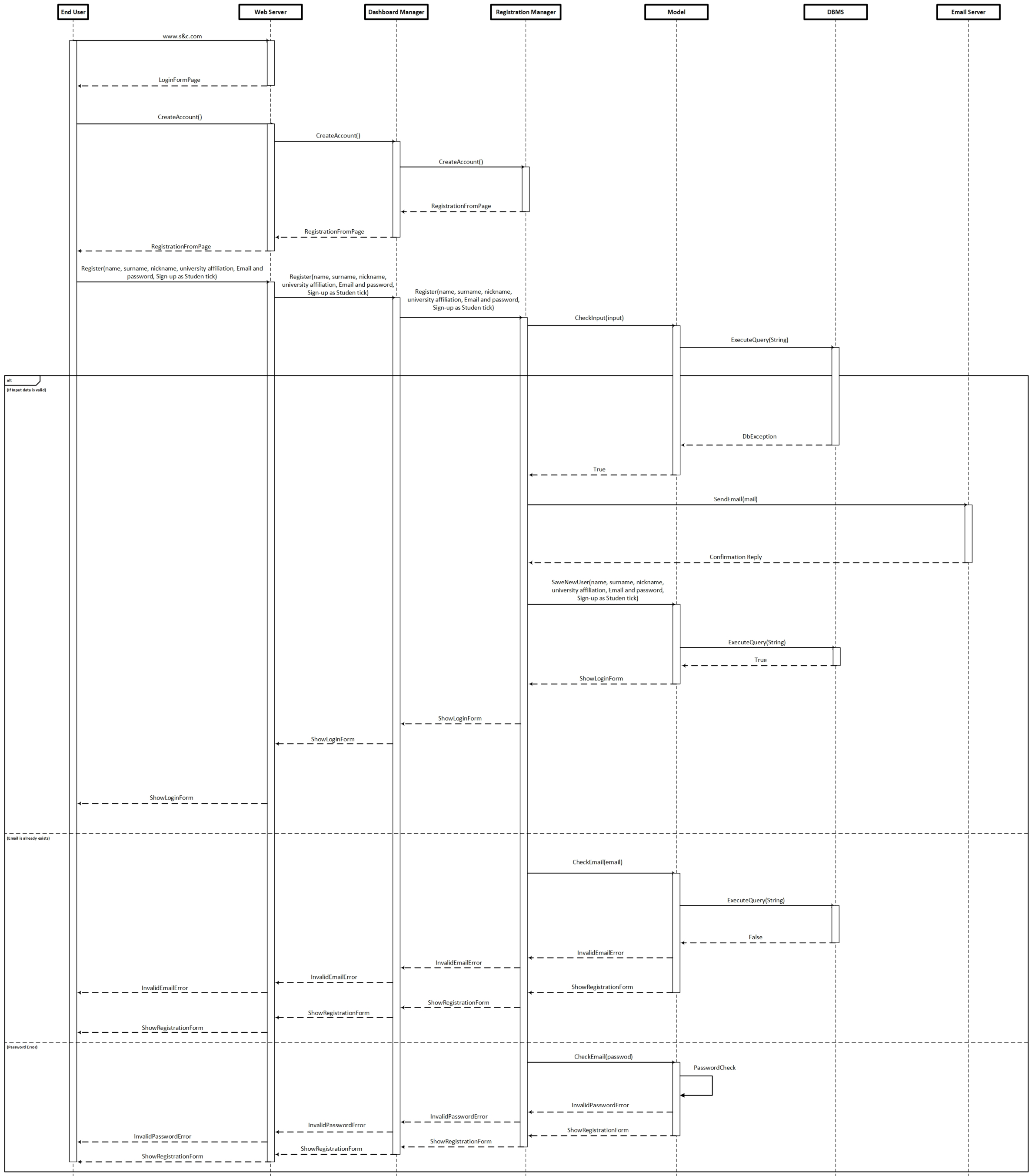

Figure 6: Student Registration Runtime View

2.4.2 Registering as Company

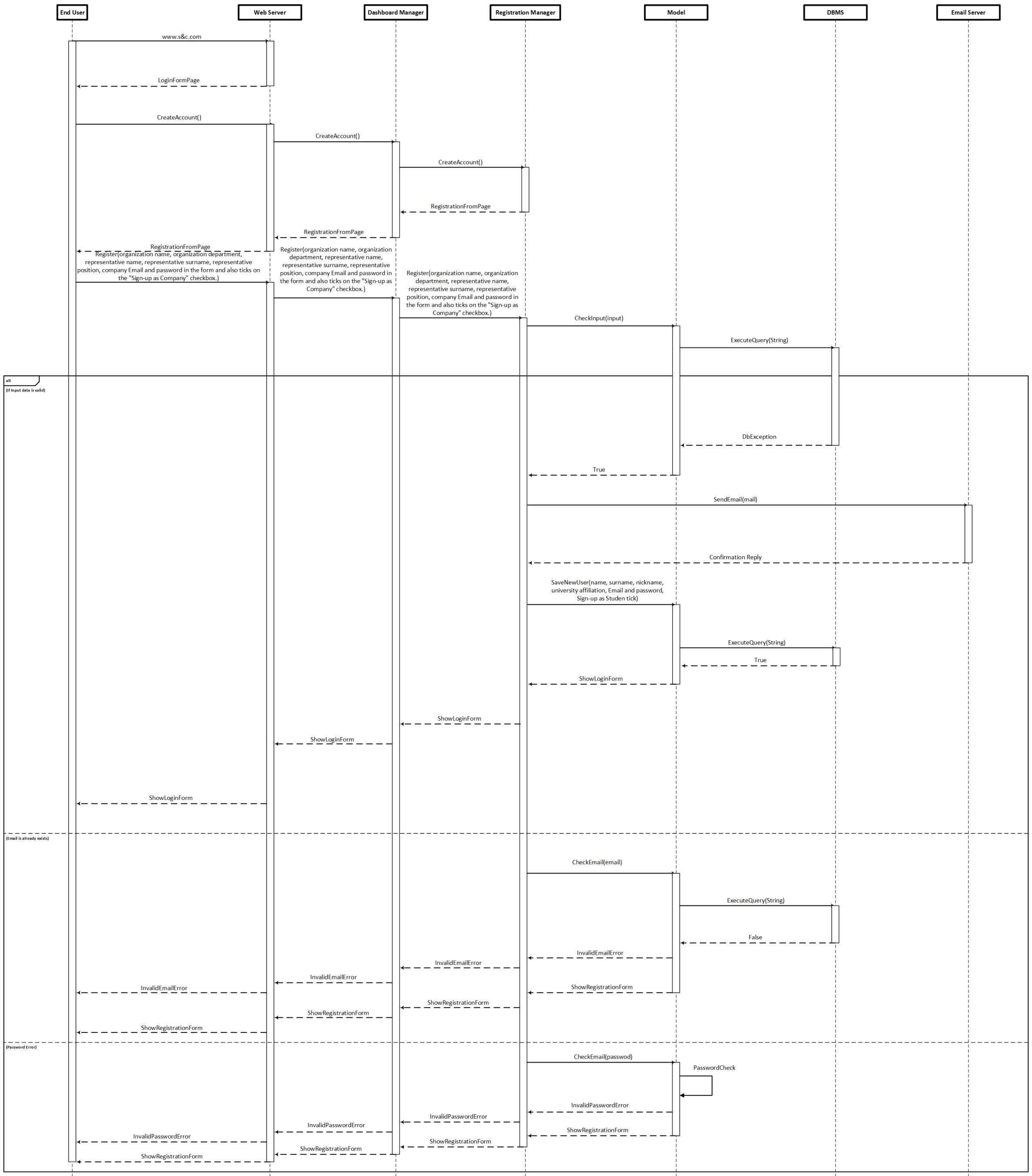

Figure 7: Company Registration Runtime View

2.4.3 Registering as University

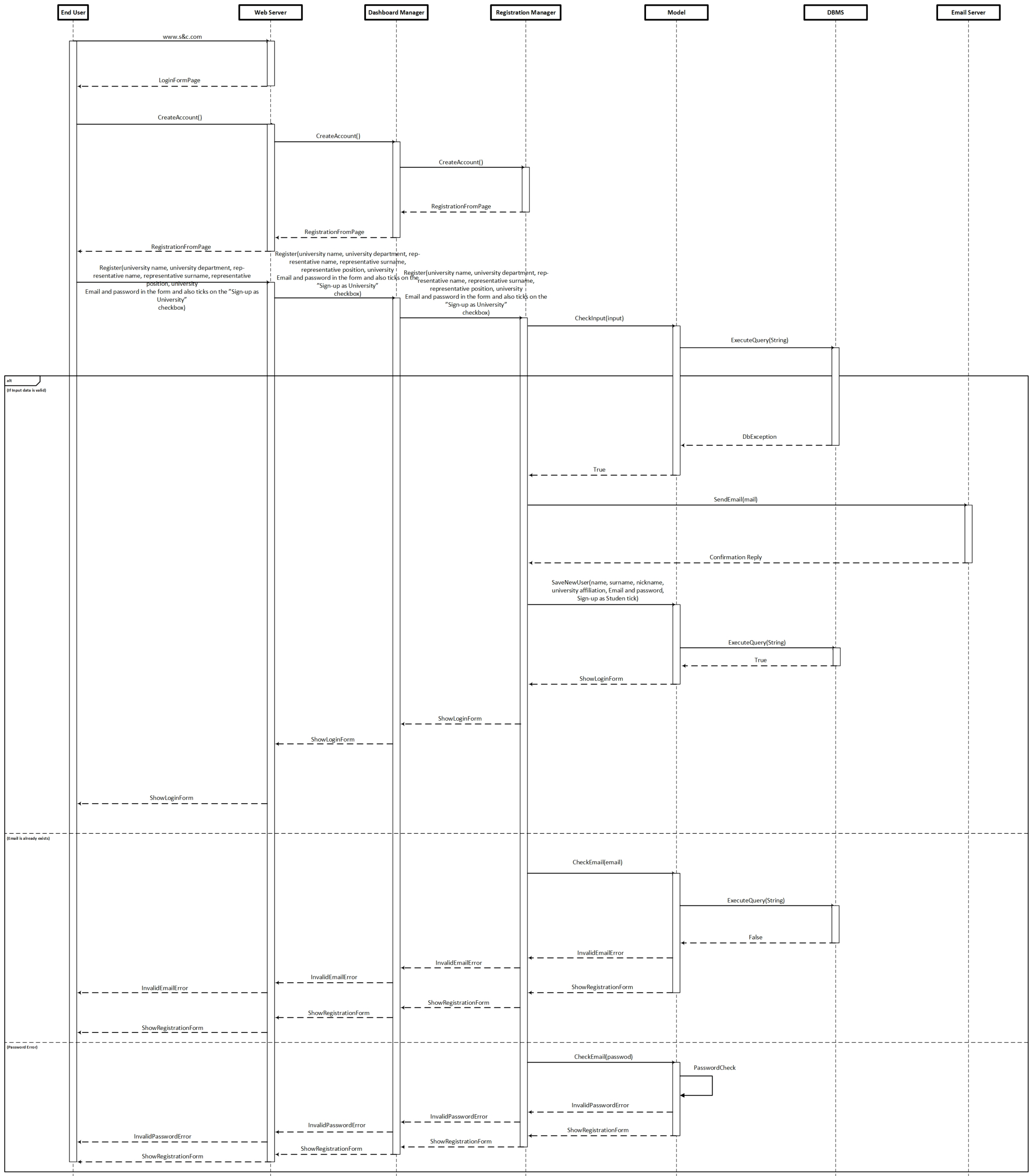

Figure 8: University Registration Runtime View

2.4.4 Login

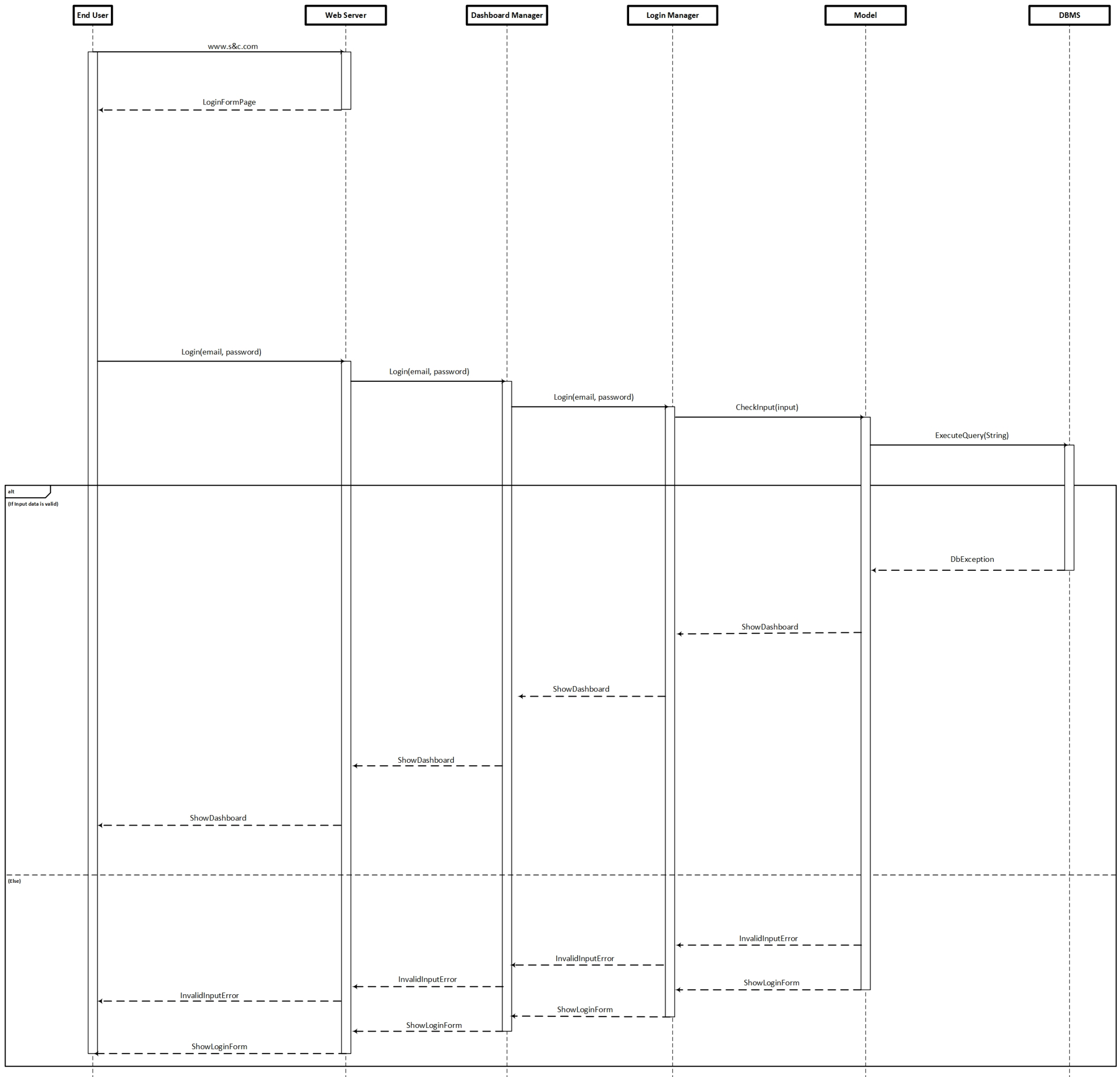

Figure 9: Login Runtime View

2.4.5 Manage Resume

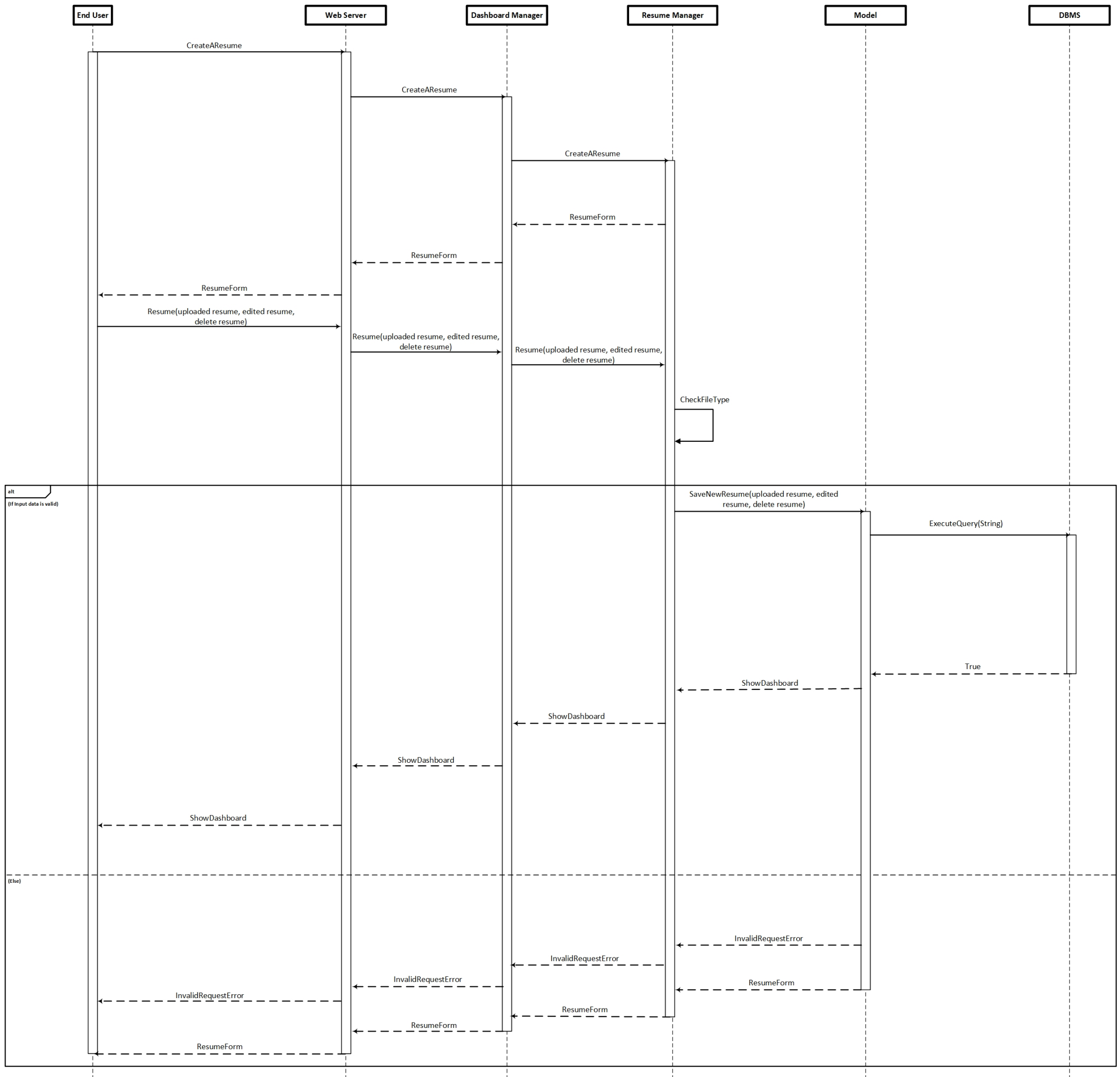

Figure 10: Manage Resume Runtime View

2.4.6 View Internship

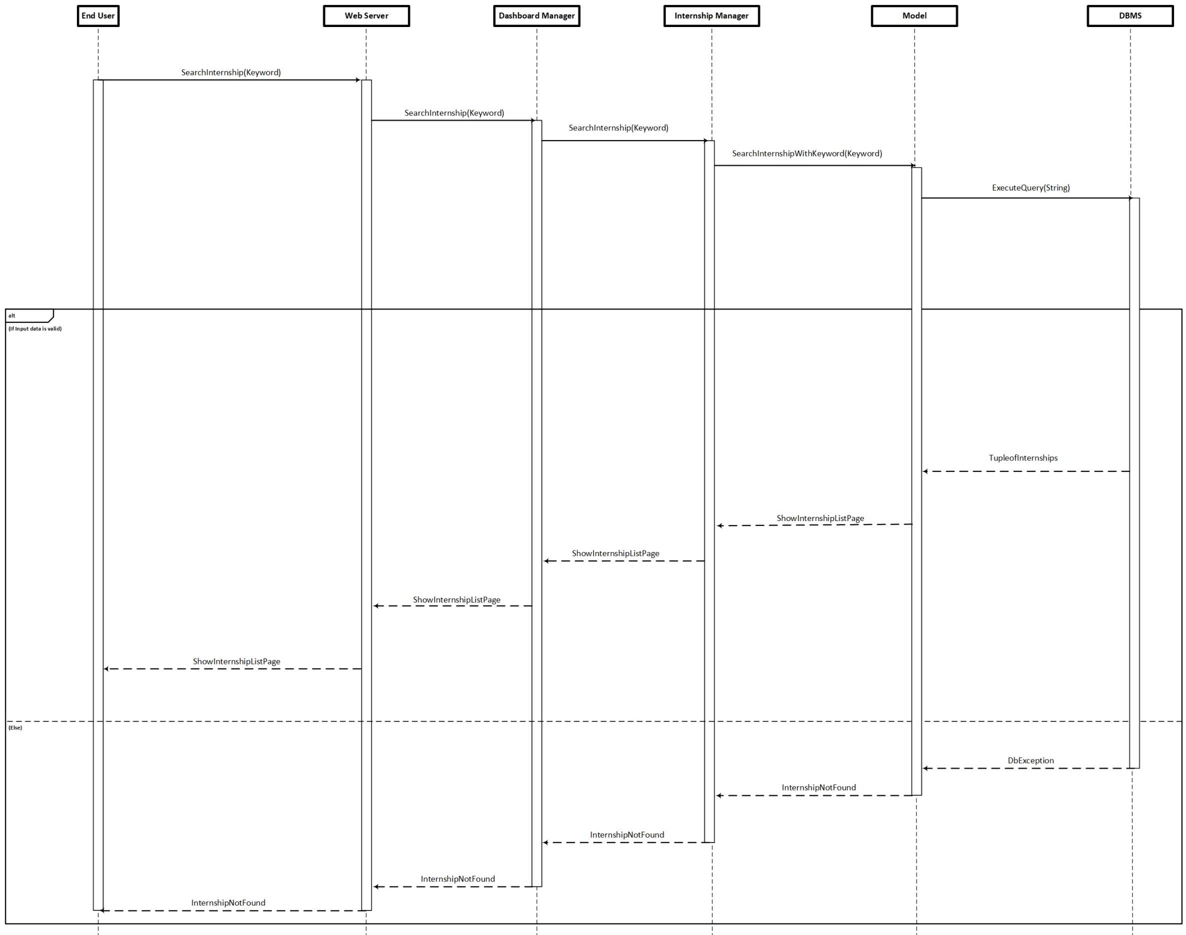

Figure 11: View Internship Runtime View

2.4.7 Sending Application

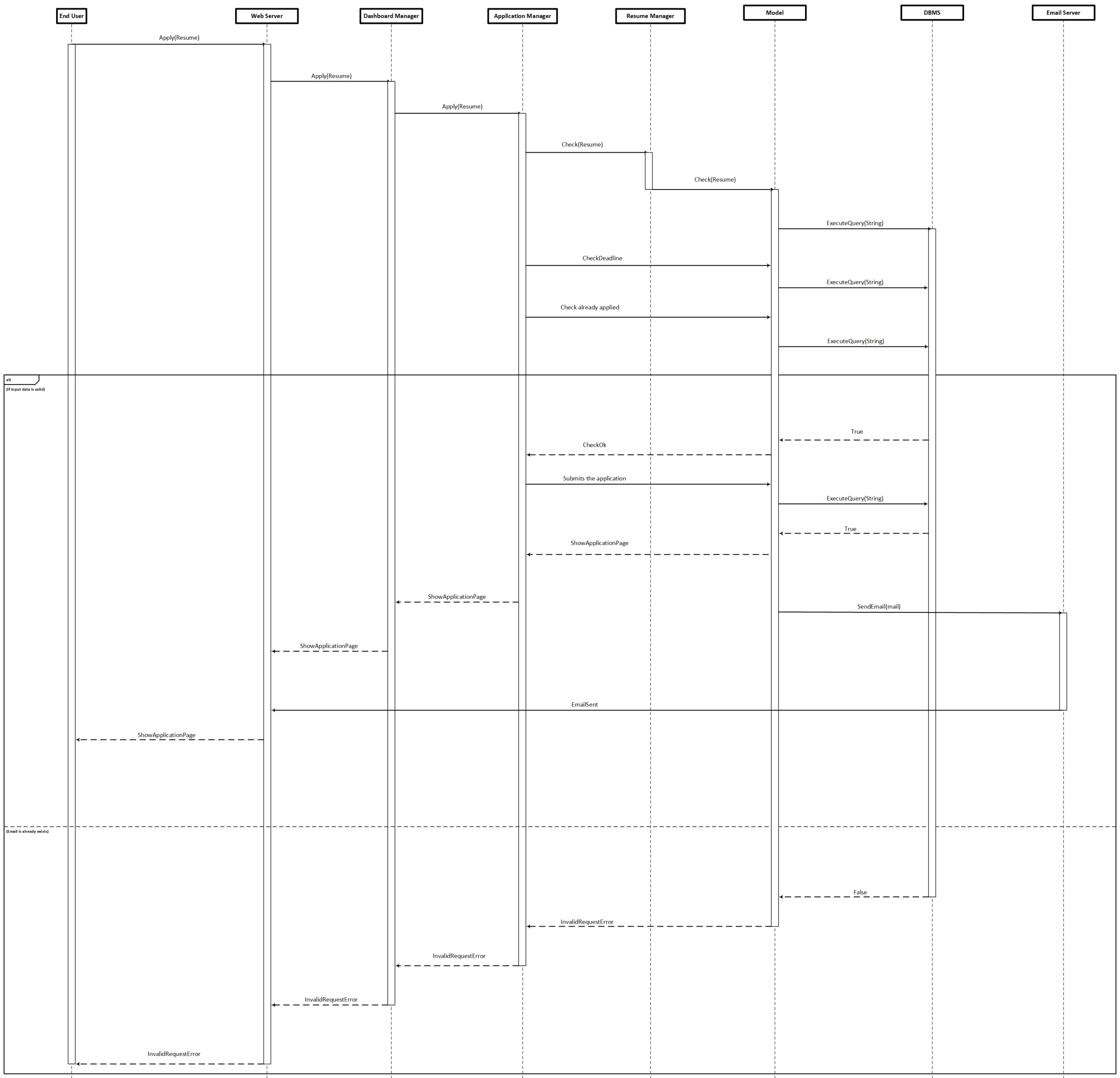

Figure 12: Sending Application Runtime View

2.4.8 Track Application

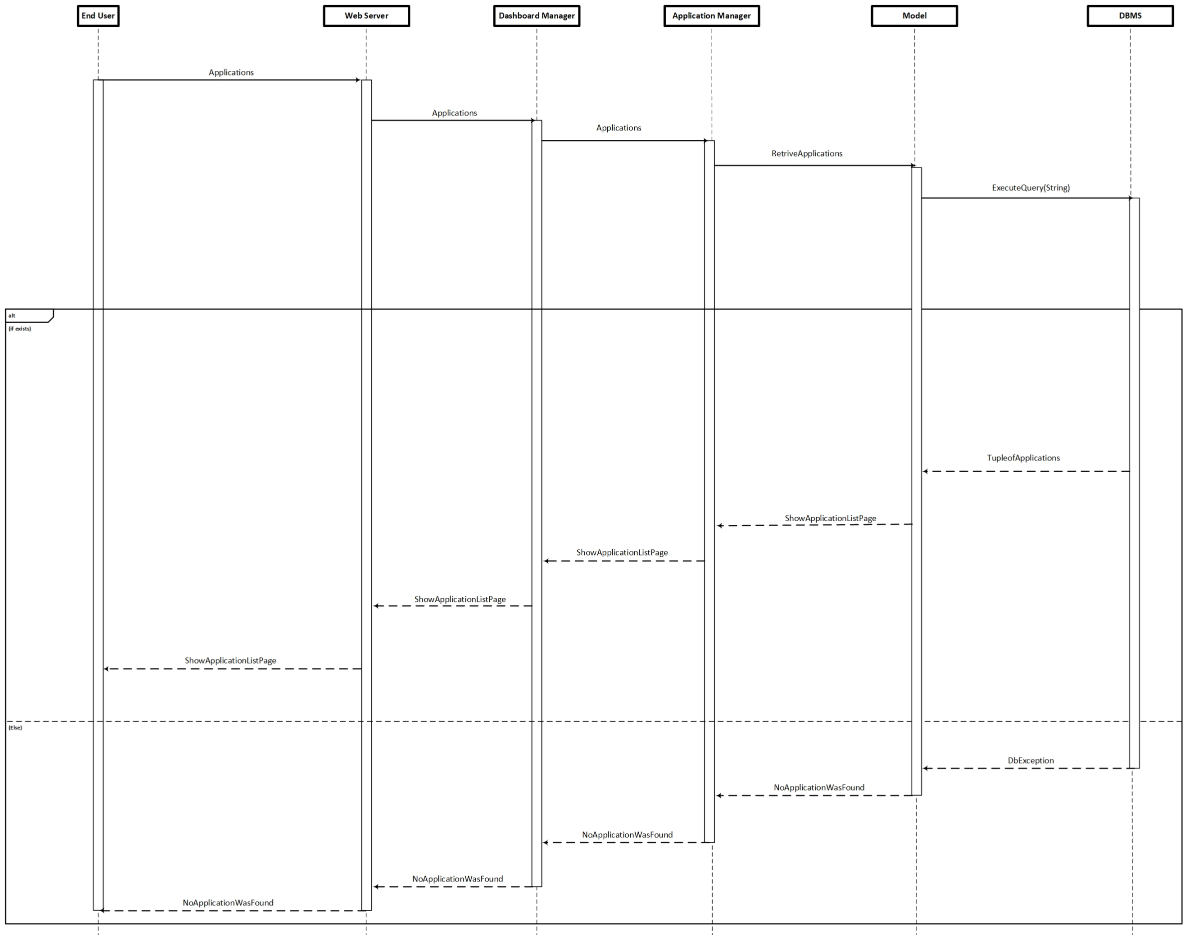

Figure 13: Track Application Runtime View

2.4.9 Complaint Submission

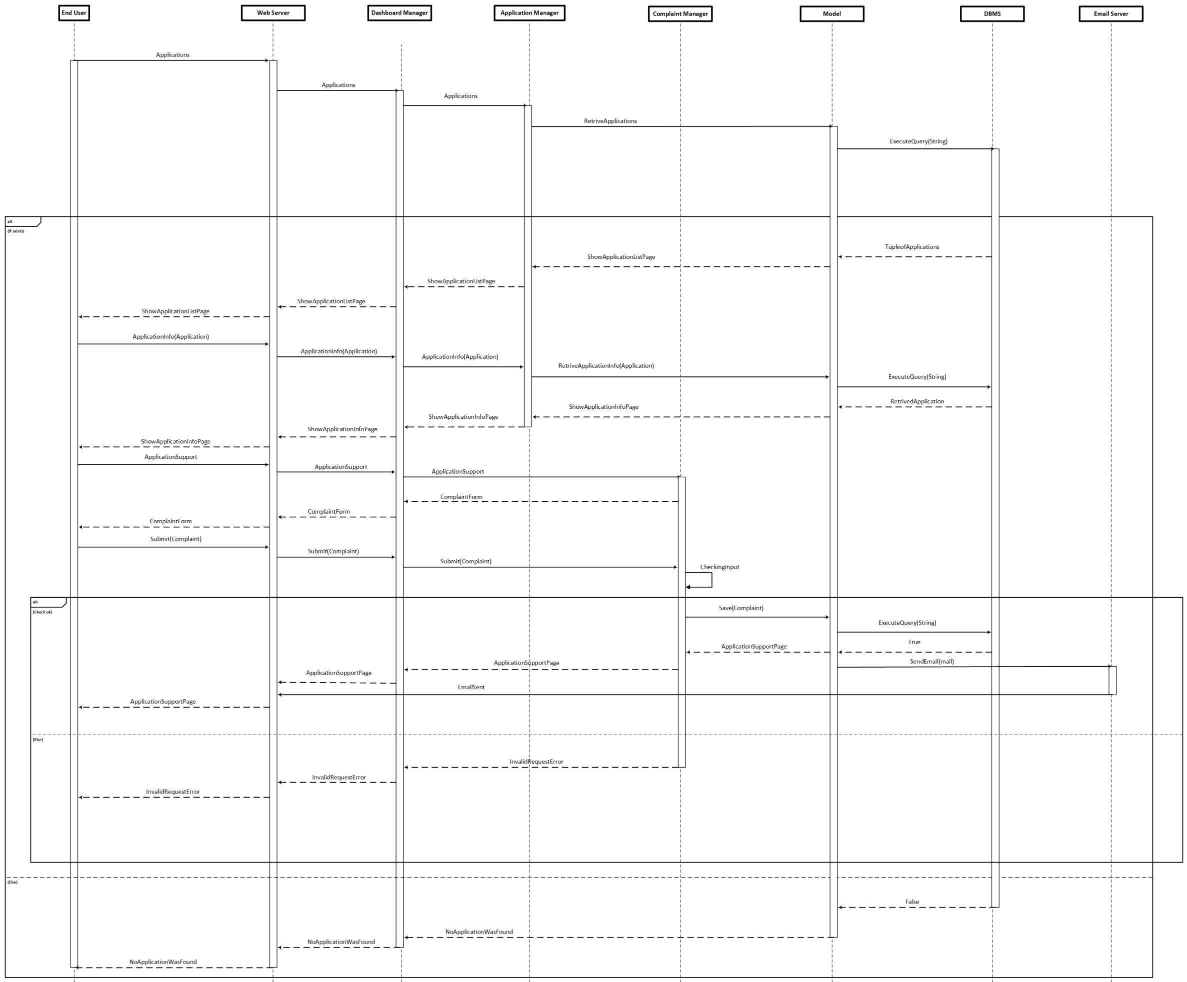

Figure 14: Complaint Submission Runtime View

2.4.10 Questionnaire Submission

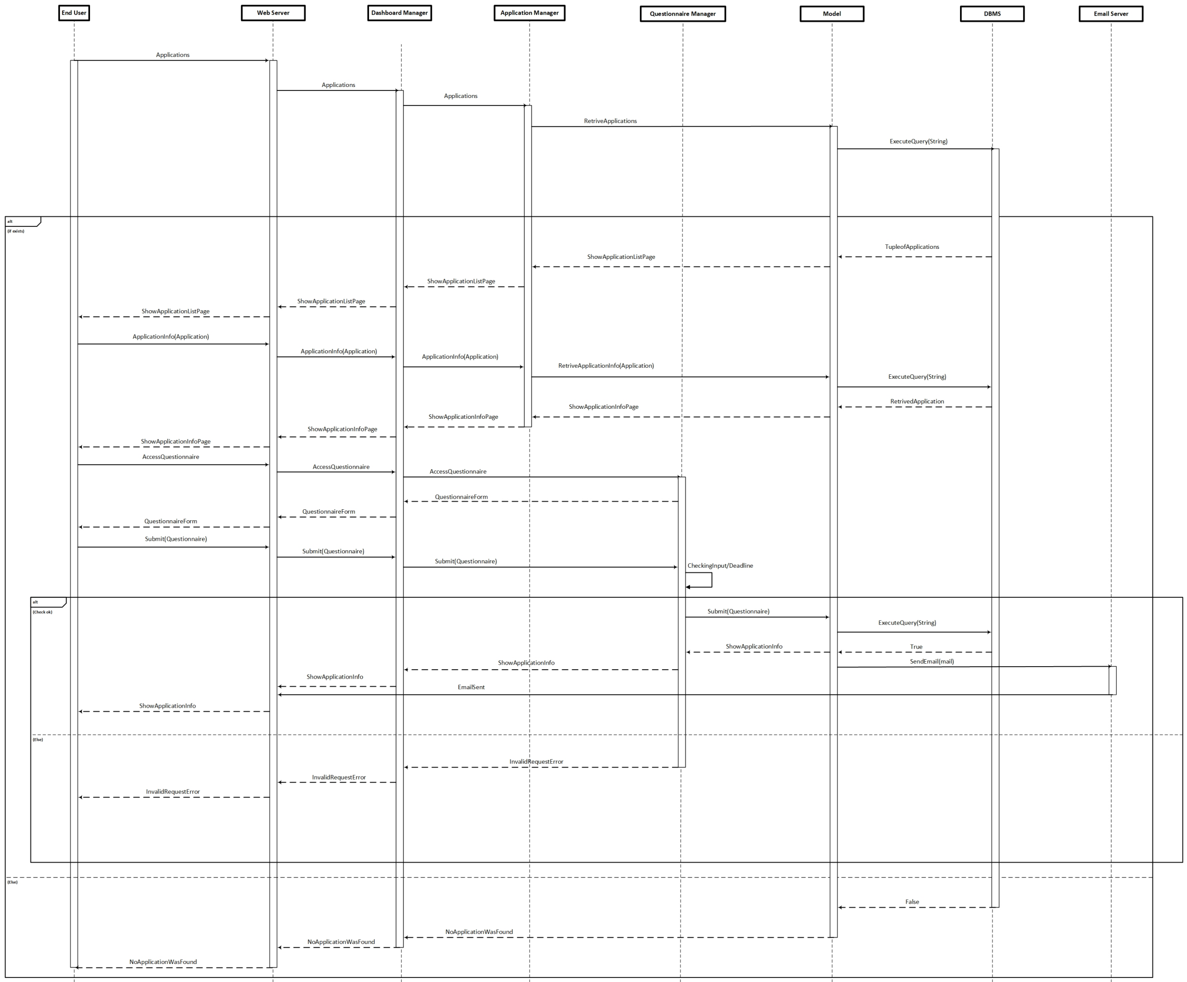

Figure 15: Questionnaire Submission Runtime View

2.4.11 Accept/Reject Offer

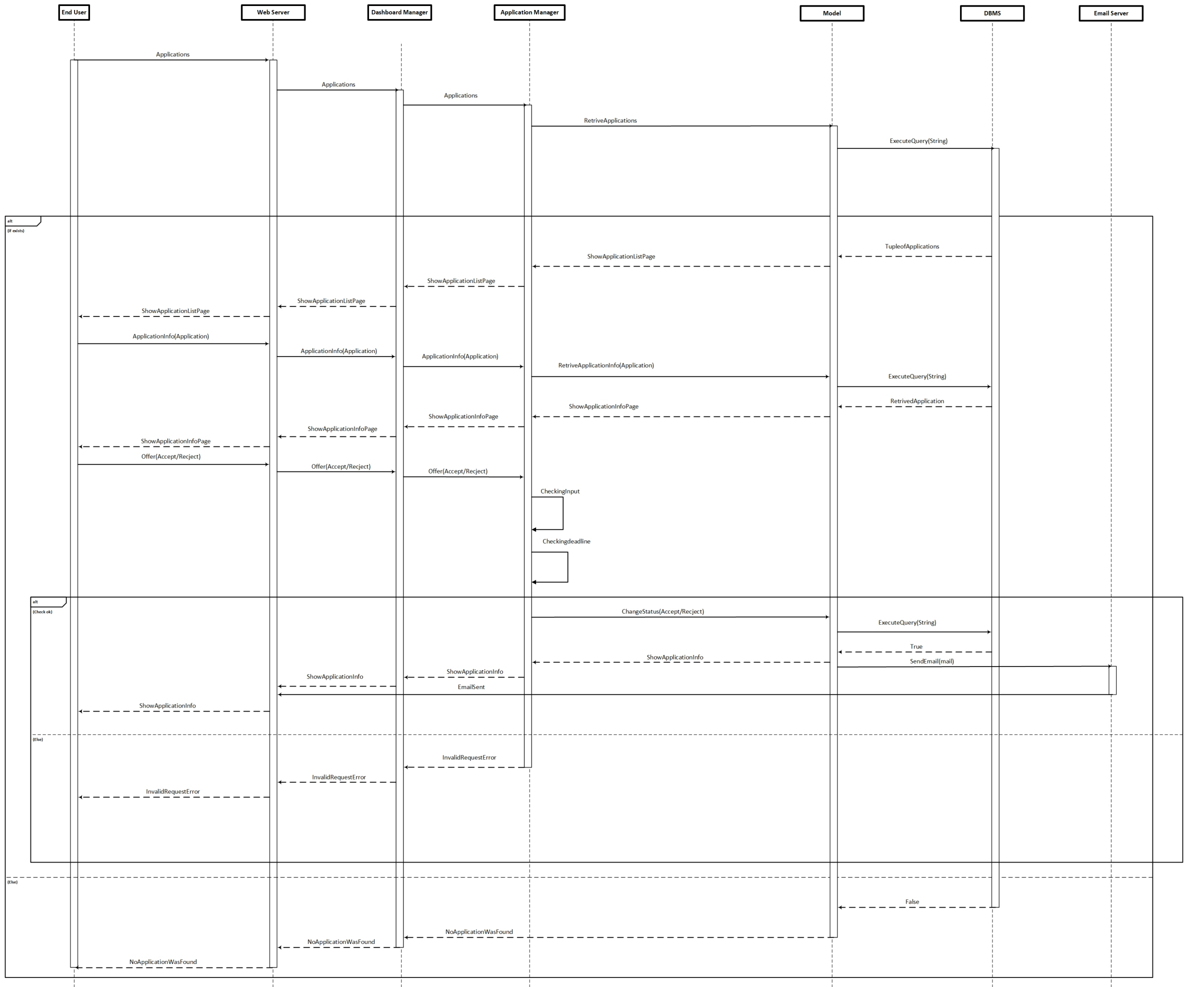

Figure 16: Accept/Reject Offer Runtime View

2.4.12 Meeting Attendance

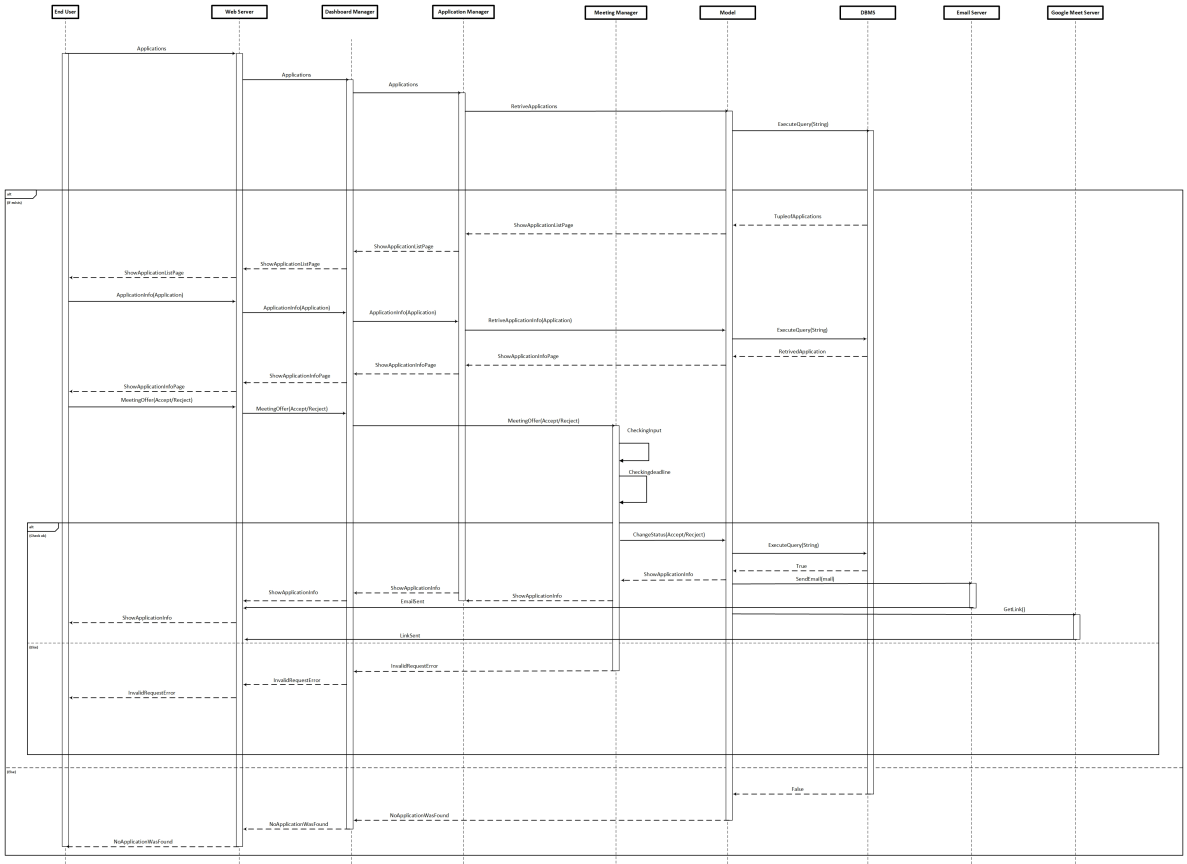

Figure 17: Accept/Reject Meeting Attendance Offer Runtime View

2.4.13 Internship Posting

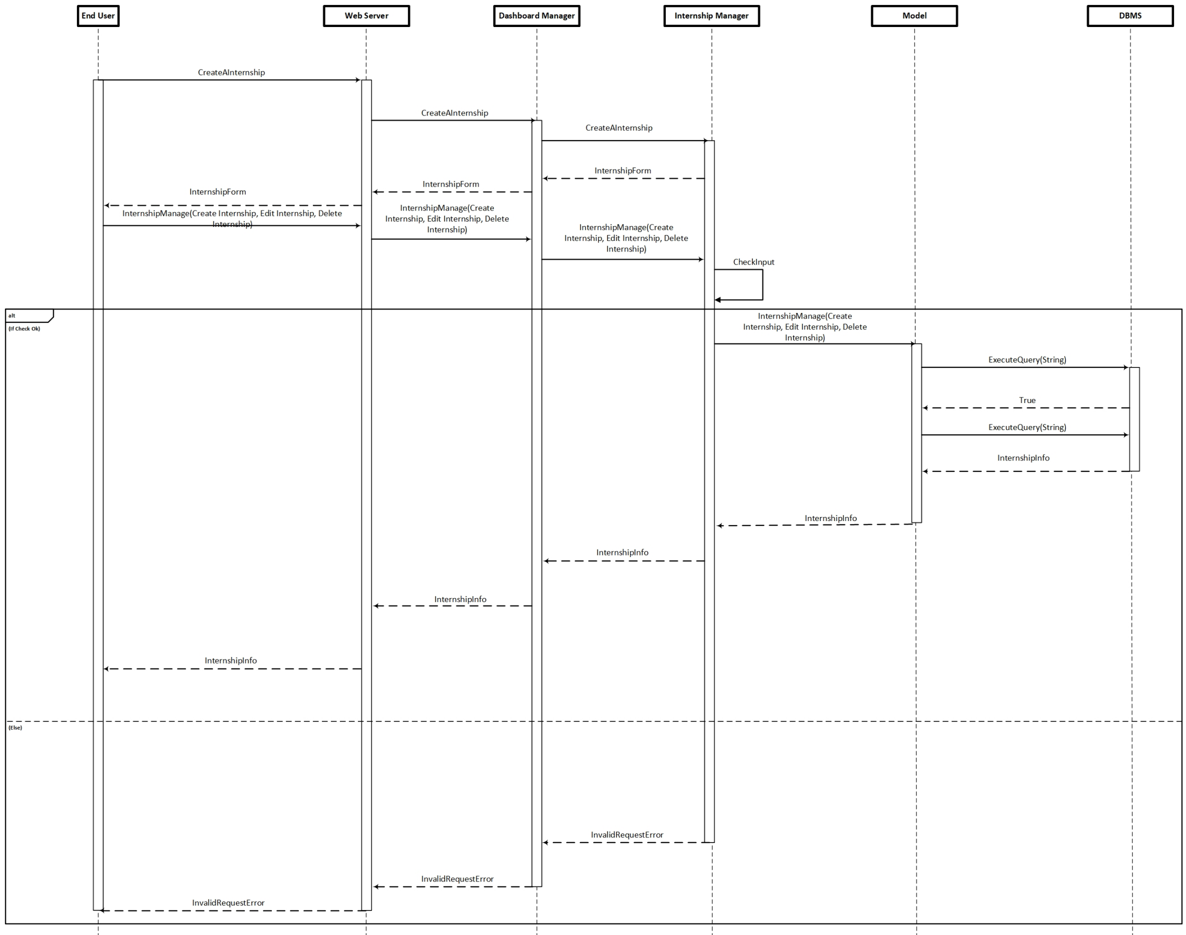

Figure 18: Internship Posting Runtime View

2.4.14 View Applicants

2.5 Component Interfaces

2.6 Selected Architectural Styles and Patterns

2.7 Other Design Decisions

2.7.1 Availability

2.7.2 Scalability

2.7.3 Notification Handling

2.7.4 Ease of Deployment

2.7.5 Data Storage

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3.1 Overview

3.2 Header

3.3 User Interfaces

4 Requirements Traceability

5 Implementation, Integration and Test Plan

6 Effort Spent

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